

# Search for $t\bar{t}$ resonances with ATLAS data

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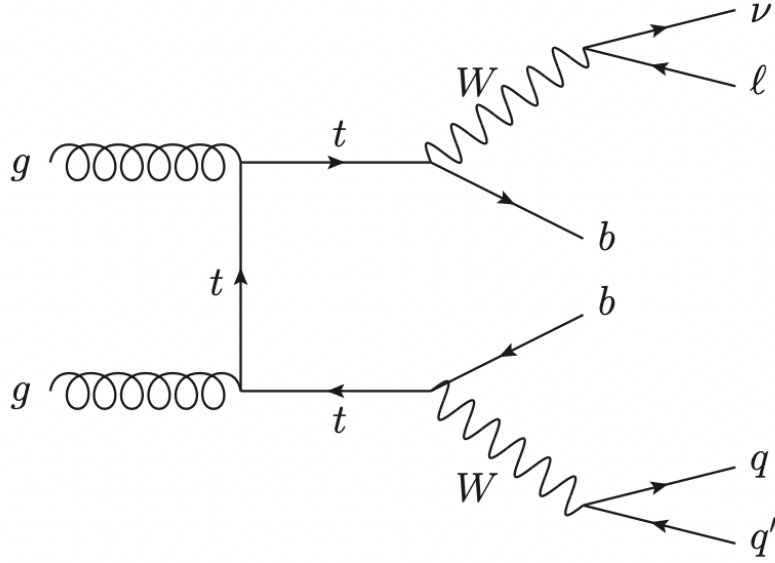
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# 1 Introduction

In this analysis, a search for a new particle decaying into a  $t\bar{t}$  pair is done using data corresponding to  $1\text{ pb}^{-1}$  collected by the ATLAS experiment at the Large Hadron Collider (LHC) at CERN. Such a particle could be associated with a resonance predicted by an expansion of the Standard Model of particle physics (SM).

## 2 Theoretic Foundations

The top quark is the heaviest elementary particle in the SM with a mass of  $173\text{ GeV}$  [2] and a correspondingly short lifetime of approximately  $5 \times 10^{-25}\text{ s}$ . Therefore, unlike all other quarks, it does not hadronise before decaying, offering a unique research sector in particle physics. In Figure 1, an example Feynman diagram of top quark production and decay is shown at leading order. Top quarks are dominantly produced in a  $t\bar{t}$  pair via the strong interaction, e.g. via gluon gluon fusion in  $pp$  collisions. The top quark decays almost exclusively via the weak interaction into a  $W^+$  boson and a  $b$  quark, but also a  $s$  or  $d$  quark is possible, although highly suppressed. The  $W^+$  boson can either decay into a lepton and its corresponding neutrino,  $W^+ \rightarrow \ell^+ \nu_\ell$ , where  $\ell \in \{e, \mu, \tau\}$ , or into a quark and an anti-quark,  $W^+ \rightarrow q\bar{q}'$ , where  $q \in \{u, c\}$  and  $\bar{q}' \in \{\bar{d}, \bar{s}, \bar{b}\}$ . Decays of the  $W^+$  boson into a quark and an anti-quark of different generations are possible, but CKM suppressed. The quarks hadronise, resulting in colour neutral jets. The channel in which both  $W$  bosons decay into leptons and neutrinos is called the *dilepton channel*, whereas the channel in which both decay into quarks is called the *all-hadronic* channel. If one  $W$  boson decays into leptons and the other one into quarks it is called the *lepton+jets* channel.



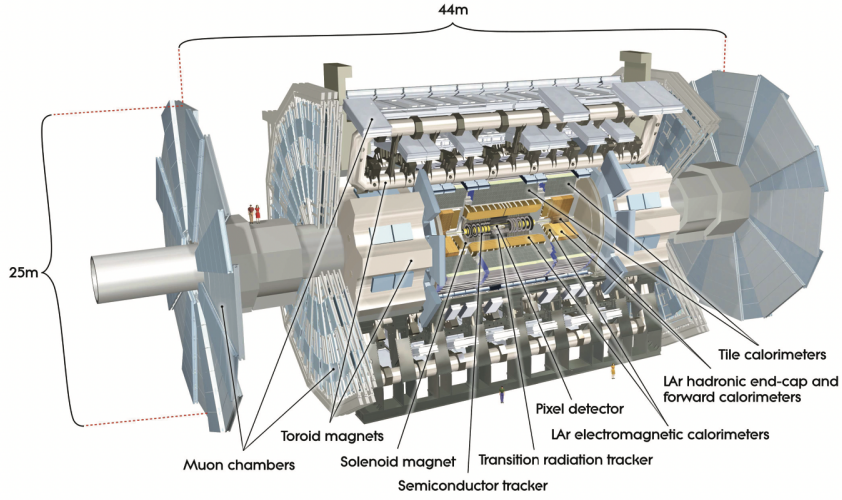
**Abbildung 1:** Example of a leading order Feynman diagram for top quark production and decay in  $pp$  collisions. The  $t\bar{t}$  pair is produced via gluon-gluon fusion and decays in the *lepton+jets* channel. [1]

### 3 The ATLAS detector

The data analysed in this analysis was collected by the ATLAS experiment [3] at the LHC [4] at CERN. The detector is depicted in Figure 2. It is designed to be a general-purpose detector, ensuring that the detector is able to detect almost any form of occurring new physics. The detector can be divided into four major subsystems: the inner detector, the calorimeters, the muon spectrometer and the magnet system.

The inner detector is the part of the detector nearest to the interaction point. It consists of a silicon pixel detector, followed by the semi-conductor tracker, which is a silicon strip detector, and the transition radiation tracker (TRT), which is a combination of a straw traker and a transition radiation tracker. The inner detector is used to measure the tracks of charged particles inside the detector. The pixel detector has a resolution high enough to identify displaced secondary vertices, which is important for identification of  $b$ -jets for example. The TRT also offers the ability to identify electrons and positrons. Due to their low mass they can reach higher velocities resulting in a larger amount of emitted transition radiation.

The inner detector is followed by the calorimeters. This is the only detector component able to detect neutral hadrons and photons. The LAr calorimeter is an electromagnetic sampling calorimeter with liquid argon as sampling material. For enhanced resolution it has an accordion-like structure. The LAr calorimeter is used to measure the energy of electrons, positrons and photons that deposit all of their energy in the calorimeter. Afterwards, surviving particles arrive the tile calorimeter which is a hadronic calorimeter.



**Abbildung 2:** Sketch of the ATLAS detector. [3]

It works the same as the LAr calorimeter, but is optimised to measure the deposited energy of hadronic showers.

The outermost system is the muon spectrometer, since muons are observed as minimum ionising particles in  $pp$  collisions and are therefore able to travel through the whole detector, without getting absorbed. The muon spectrometer is used to measure tracks of those muons, which can be matched to tracks in the inner detector.

The inner detector is surrounded by a solenoid magnet. The magnetic field inside the inner detector causes charged particles to drift. From the resulting curvature of the tracks, a particle's momentum can be inferred. Similarly, a toroid magnet generates a magnetic field in the muon chambers, allowing to infer muon momenta the same way.

The ATLAS detector is not able to detect neutrinos. These only interact with matter exclusively via the weak interaction and thus typically pass through normal matter undetected.

## 4 Analysis strategy

In this analysis, a search for  $t\bar{t}$  resonances, caused by a hypothetical  $Z'$  particle that decays into a  $t\bar{t}$  pair, is performed. Since the *all-hadronic* channel has a very large background contribution and the *dilepton* channel has a very small branching fraction, only the *lepton+jets* channel is considered in this analysis. To account for neutrinos from the  $W$  decays, missing transverse momentum is required. To distinguish between  $b$  and light-hadron-jets,  $b$ -tagging is used. For the distinction a displaced secondary vertex is important among other criteria. An event selection is applied using these and additional criteria to increase the signal-to-background ratio. To model the background contribution, Monte Carlo (MC) simulations are used. To check the validity of the MC simulations, these are compared to data. In the end, a final discriminant is chosen and used in a

statistical analysis to set limits on the cross sections of  $Z'$  particles at different masses.

## 5 Analysis

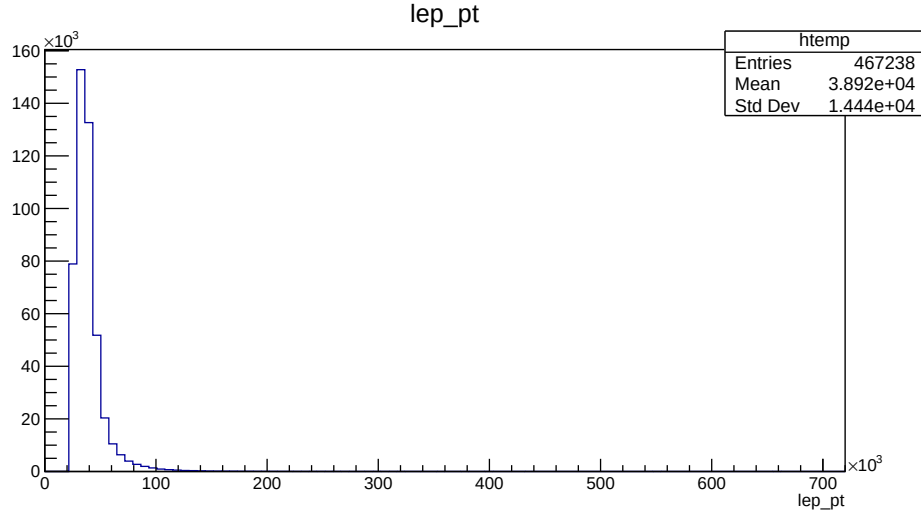
### 5.1 The samples

A preselection was done to the data used in the analysis. Events are required to either include a muon or an electron. In the preselection, a cut on both, the jet transverse momentum, `jet_pt`, and the lepton transverse momentum, `lep_pt`, of more than 25 GeV was applied. Furthermore, selected jets and leptons are required to have a pseudorapidity of  $|\eta| < 2.5$ . The lepton transverse momentum, shown in Figure 3, rises sharply at low momenta and has a peak around 50 GeV. After that it falls exponentially.

### 5.2 The event selection

To increase the signal to background ratio, a selection is applied which specifically targets the final state of the `lepton+jets` channel of the signal process. The final state consists of one charged lepton, a neutrino, two bottom quarks and two other quarks from the decay of a  $W$  boson. The selection consists of 7 cuts, which are applied subsequently. The efficiencies of the selection ( $A \cdot \epsilon = \text{\#events(after cut)}/\text{\#events(before selection)}$ ) of each cut and that of the combination of all cuts are listed in Tables 1 and 2 for all Monte Carlo samples, which include background simulations as well as  $Z'$  simulations assuming different masses.

For the jets, the number of good jets in one event is required to be  $\geq 4$ , while exactly one lepton is required in the events. Since the final state contains a neutrino, missing transverse energy  $met_{et} > 40$  GeV is required. To suppress combinatorial background, the lepton is required to have a transverse momentum  $lep\_pt > 50$  GeV and one of the jets has to pass  $jet\_pt > 80$  GeV. To calculate the efficiency of the cut on the lepton transverse momentum only, at least one with  $lep\_pt > 50$  GeV is required. The number of b-tagged jets in one event has to be  $\geq 2$  and one of the b-jets has to fulfill  $jet\_pt > 50$  GeV.



**Abbildung 3:** Histogram of the transverse momentum of the leptons in one of the data samples created with the **TBrowser** environment from **Root**. In addition, the histogram contains the mean value and the standard deviation of the variable **lep\_pt** while the unit of the transverse momentum is MeV.

**Tabelle 1:** The efficiency of the selection of the first five cuts for all simulated samples.

| sample             | #good jets | #leptons | $p_T$ (missing) | $p_T$ (lepton) | $p_T$ (jet) |
|--------------------|------------|----------|-----------------|----------------|-------------|
| diboson.el.root    | 0.0202309  | 0.90444  | 0.538132        | 0.447311       | 0.15773     |
| diboson.mu.root    | 0.016195   | 0.873096 | 0.562105        | 0.439021       | 0.145303    |
| singletop.el.root  | 0.0906053  | 0.980647 | 0.642119        | 0.447838       | 0.361111    |
| singletop.mu.root  | 0.0890013  | 0.978754 | 0.657748        | 0.42671        | 0.355983    |
| ttbar.el.root      | 0.436571   | 0.931562 | 0.730775        | 0.559994       | 0.628057    |
| ttbar.mu.root      | 0.434872   | 0.92004  | 0.735454        | 0.542928       | 0.625919    |
| wjets.el.root      | 0.00489397 | 0.999992 | 0.464057        | 0.170926       | 0.0521672   |
| wjets.mu.root      | 0.00482915 | 0.999995 | 0.508766        | 0.158871       | 0.0481682   |
| zjets.el.root      | 0.0118665  | 0.716686 | 0.17578         | 0.294029       | 0.0688467   |
| zjets.mu.root      | 0.00627221 | 0.501521 | 0.234238        | 0.323116       | 0.0557154   |
| zprime1000.el.root | 0.594403   | 0.933907 | 0.865877        | 0.796663       | 0.942842    |
| zprime1000.mu.root | 0.609297   | 0.918215 | 0.860023        | 0.782111       | 0.942072    |
| zprime1250.el.root | 0.614838   | 0.935476 | 0.888074        | 0.809224       | 0.958304    |
| zprime1250.mu.root | 0.630988   | 0.924588 | 0.884543        | 0.803705       | 0.960049    |
| zprime1500.el.root | 0.630429   | 0.940084 | 0.906332        | 0.828493       | 0.966117    |
| zprime1500.mu.root | 0.639837   | 0.928664 | 0.905582        | 0.816934       | 0.969157    |
| zprime1750.el.root | 0.637424   | 0.947275 | 0.915079        | 0.825727       | 0.97061     |
| zprime1750.mu.root | 0.642951   | 0.938331 | 0.917956        | 0.816627       | 0.971889    |
| zprime2000.el.root | 0.636319   | 0.945145 | 0.919928        | 0.815458       | 0.969052    |
| zprime2000.mu.root | 0.659439   | 0.940585 | 0.923095        | 0.817801       | 0.972448    |
| zprime2250.el.root | 0.63421    | 0.94502  | 0.917993        | 0.797853       | 0.969456    |
| zprime2250.mu.root | 0.652435   | 0.944785 | 0.923501        | 0.815951       | 0.968574    |
| zprime2500.el.root | 0.636889   | 0.948017 | 0.912399        | 0.799576       | 0.960916    |
| zprime2500.mu.root | 0.658653   | 0.943286 | 0.919777        | 0.81299        | 0.95949     |
| zprime3000.el.root | 0.6        | 0.943673 | 0.893054        | 0.766698       | 0.924833    |
| zprime3000.mu.root | 0.626332   | 0.939849 | 0.897084        | 0.786596       | 0.942933    |
| zprime400.el.root  | 0.336048   | 0.930846 | 0.701021        | 0.495382       | 0.485051    |
| zprime400.mu.root  | 0.339518   | 0.920131 | 0.70503         | 0.477329       | 0.492807    |
| zprime500.el.root  | 0.418102   | 0.932419 | 0.741854        | 0.617663       | 0.725946    |
| zprime500.mu.root  | 0.401054   | 0.922425 | 0.747411        | 0.59904        | 0.716249    |
| zprime750.el.root  | 0.537326   | 0.923685 | 0.820961        | 0.740798       | 0.899251    |
| zprime750.mu.root  | 0.532581   | 0.913321 | 0.831628        | 0.722732       | 0.89058     |

**Tabelle 2:** The efficiency of the selection of the sixth and seventh cut and for the application of all mentioned cuts for all simulated samples.

| sample             | #b-tagged jets | $p_T$ (b-tagged jets) | all cuts    |
|--------------------|----------------|-----------------------|-------------|
| diboson.el.root    | 0.00458489     | 0.0319561             | 0.00011688  |
| diboson.mu.root    | 0.00473332     | 0.0308818             | 9.56227e-05 |
| singletop.el.root  | 0.112618       | 0.43978               | 0.00795335  |
| singletop.mu.root  | 0.111103       | 0.434325              | 0.00747094  |
| ttbar.el.root      | 0.320864       | 0.629536              | 0.0505078   |
| ttbar.mu.root      | 0.320096       | 0.630451              | 0.0482129   |
| wjets.el.root      | 0.000225222    | 0.0044388             | 1.42213e-05 |
| wjets.mu.root      | 0.000230063    | 0.0040673             | 1.52894e-05 |
| zjets.el.root      | 0.00121684     | 0.0140795             | 5.25442e-05 |
| zjets.mu.root      | 0.0011521      | 0.0128361             | 3.3859e-05  |
| zprime400.el.root  | 0.284152       | 0.574745              | 0.0226057   |
| zprime400.mu.root  | 0.282072       | 0.571386              | 0.0222244   |
| zprime500.el.root  | 0.316292       | 0.646188              | 0.0502468   |
| zprime500.mu.root  | 0.308511       | 0.645641              | 0.0480136   |
| zprime750.el.root  | 0.365461       | 0.740175              | 0.121439    |
| zprime750.mu.root  | 0.36657        | 0.741888              | 0.124814    |
| zprime1000.el.root | 0.398493       | 0.773843              | 0.177395    |
| zprime1000.mu.root | 0.394313       | 0.782375              | 0.166124    |
| zprime1250.el.root | 0.394596       | 0.783834              | 0.18798     |
| zprime1250.mu.root | 0.405127       | 0.791542              | 0.190494    |
| zprime1500.el.root | 0.380037       | 0.778257              | 0.185767    |
| zprime1500.mu.root | 0.378967       | 0.781713              | 0.18635     |
| zprime1750.el.root | 0.36228        | 0.774922              | 0.189632    |
| zprime1750.mu.root | 0.363696       | 0.774679              | 0.187623    |
| zprime2000.el.root | 0.336499       | 0.752743              | 0.17177     |
| zprime2000.mu.root | 0.339962       | 0.761859              | 0.179085    |
| zprime2250.el.root | 0.321362       | 0.731581              | 0.16716     |
| zprime2250.mu.root | 0.337924       | 0.755478              | 0.178165    |
| zprime2500.el.root | 0.313824       | 0.723527              | 0.157682    |
| zprime2500.mu.root | 0.326338       | 0.743259              | 0.172799    |
| zprime3000.el.root | 0.323692       | 0.709991              | 0.145005    |
| zprime3000.mu.root | 0.327257       | 0.734997              | 0.15816     |

### 5.3 Fundamental distributions

The main background after the selection is  $t\bar{t}$ -production since it has the same final state as the signal, leading to a large selection efficiency in Table 2 compared to the other simulated SM processes. Therefore, the  $t\bar{t}$ -sample is investigated by creating histograms of several kinematic variables for the electron- and muon-samples. The following distributions were examined:

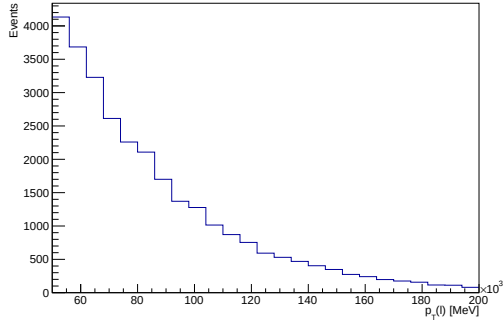


- the lepton  $p_T$ ,  $\eta$ ,  $\phi$  and  $E$ ,
- $p_T$ ,  $\eta$ ,  $\phi$  and  $E$  of all good jets,
- the number of good jets in each event,
- $p_T$ ,  $\eta$ ,  $\phi$  and  $E$  of the jets with the largest  $p_T$  for each event,
- the number of b-tagged jets,
- the magnitude of the missing transverse momentum,

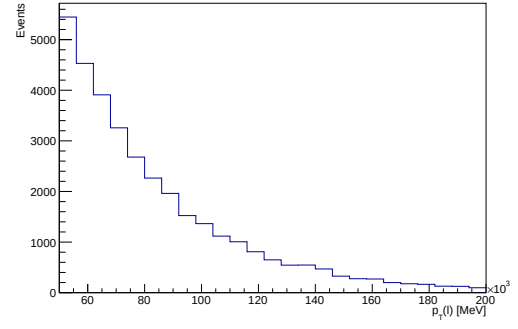
where  $p_T$  refers to the transverse momentum,  $\eta$  to the pseudorapidity,  $\phi$  to the azimuthal angle and  $E$  to the energy. The most important distributions are shown in Figures 4, 5 and 6 for the simulated  $t\bar{t}$ -sample.

Both, the missing transverse momentum and the transverse momentum shown in Figure 4, fall exponentially and appear very similar. This agrees with the expectation that the missing transverse momentum should in magnitude be roughly the same since the missing transverse momentum should be dominated by the momentum of the neutrino from the  $W$ -decay.

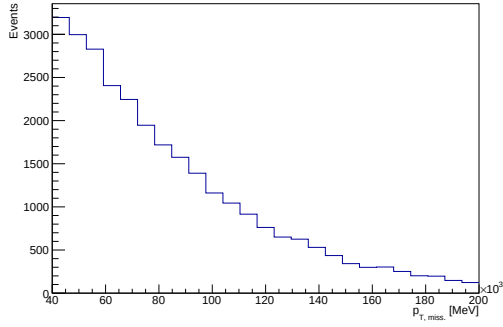
Both the number of good jets in an event and the number of b-tagged jets have a very similar distribution for the electron- and muon data samples. Since quarks can produce more than one jet in data due to initial and final state radiation or wrong reconstruction of the jets, the observation that some events contain more than four jets matches with the expectation. Since the additional jets can also originate from  $b$  quarks, also the number of  $b$ -tagged jets is expected to be larger than two in some events. additionally, it is possible that one of the quarks from the hadronic decay of the  $W$ -boson is a bottom quark, although highly suppressed. It is also possible that  $c$ -jets (and very rarely light-hadron-jets) are misreconstructed as  $b$ -jets, resulting in more than two  $b$ -tagged jets an event. The same is observed for the kinematic quantities of the jet with the largest transverse momentum of each event. All considered variables follow the expected distribution.



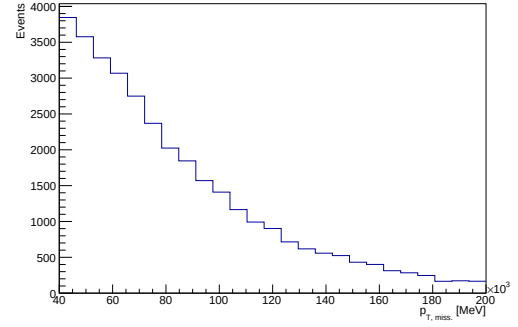
(a) Transverse momentum of the lepton in events where the lepton was tagged as electron.



(b) Transverse momentum of the lepton in events where the lepton was tagged as muon.

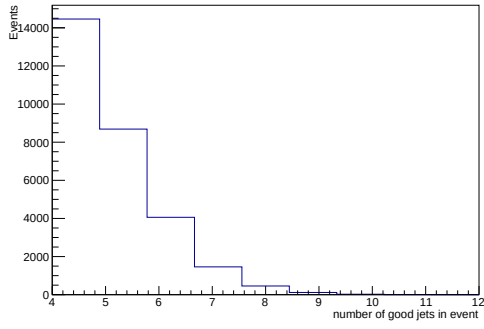


(c) Missing transverse energy of events where the lepton was tagged as electron.

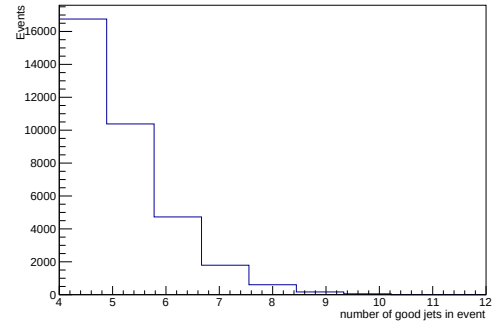


(d) Missing transverse energy of events where the lepton was tagged as muon.

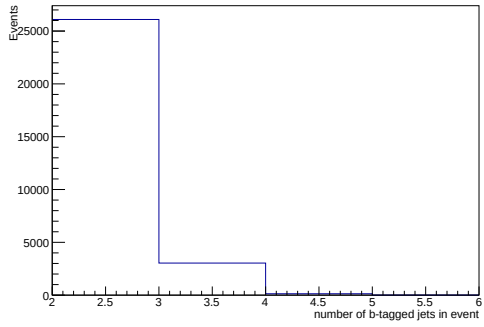
**Abbildung 4:** Transverse momentum of the lepton and missing transverse energy of each event.



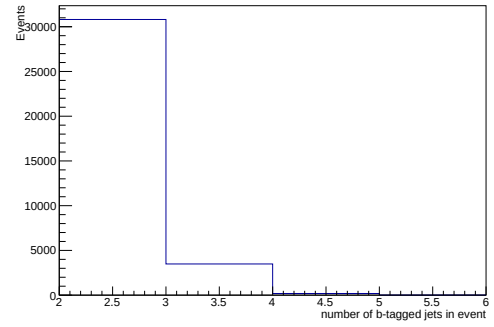
(a) Number of good jets for events where the lepton was tagged as electron.



(b) Number of good jets for events where the lepton was tagged as muon.

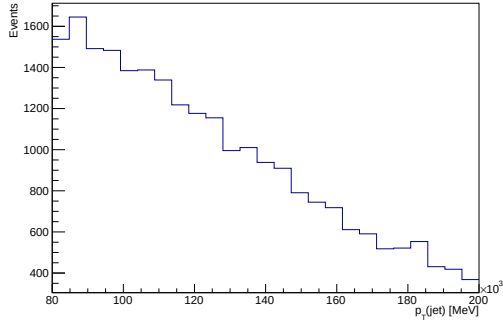


(c) Number of b-tagged jets for events where the lepton was tagged as electron.

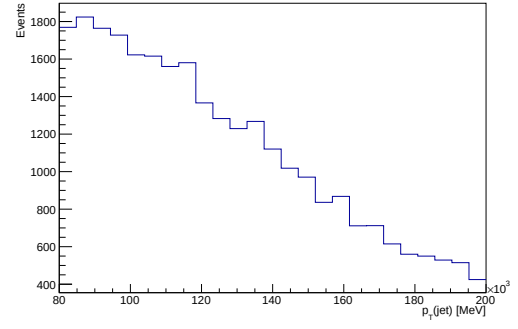


(d) Number of b-tagged jets for events where the lepton was tagged as muon.

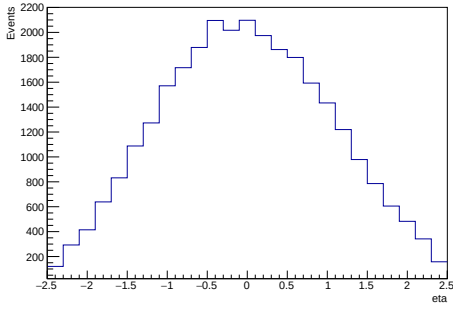
**Abbildung 5:** Number of good jets and b-tagged jets in events from the simulated  $t\bar{t}$ -sample.



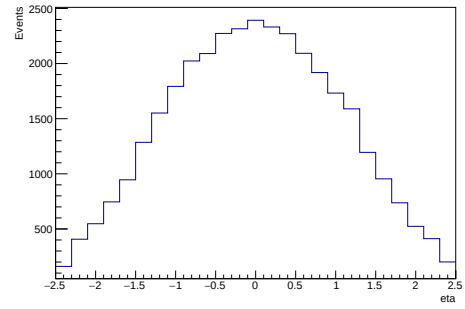
(a) Transverse momentum, lepton is electron.



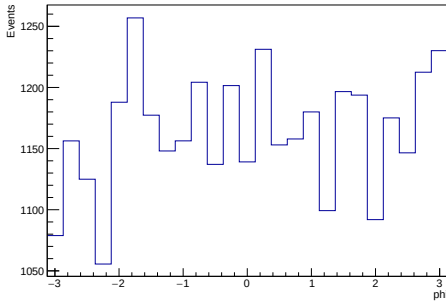
(b) Transverse momentum, lepton is muon.



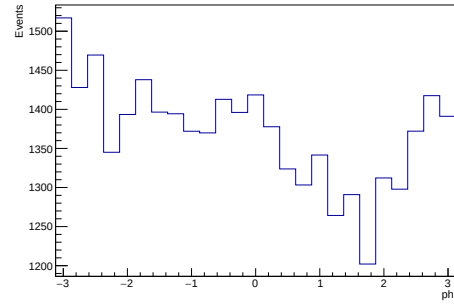
(c) Pseudorapidity, lepton is electron.



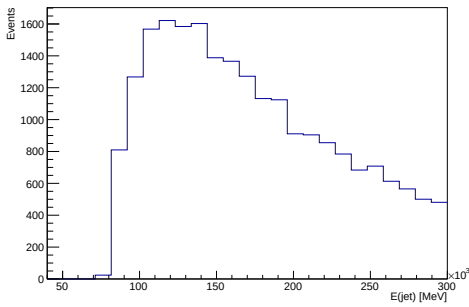
(d) Pseudorapidity, lepton is muon.



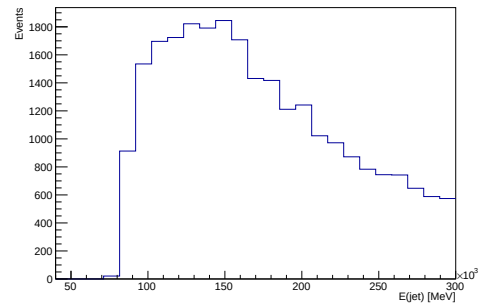
(e) Polar angle, lepton is electron.



(f) Polar angle, lepton is muon.



(g) Energy, lepton is electron.



(h) Energy, lepton is muon.

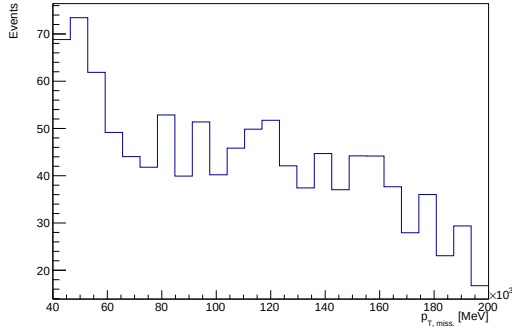
**Abbildung 6:** Histograms of the parameters for the jet with the highest transverse momentum for each event for the simulated  $t\bar{t}$ -samples.

## 5.4 Discriminant

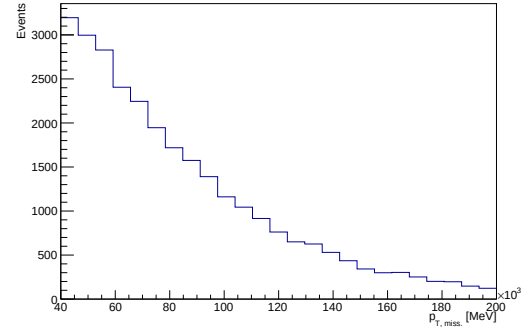
To further improve signal-to-background ratio of the samples, discriminants constructed from the four-vectors of the particles in the events are investigated. Specifically

- the missing transverse momentum ( $p_{T, \text{miss}}$ ),
- the difference of the azimuthal angle between the missing transverse momentum and the lepton where the smaller angle of  $|\phi_1 - \phi_2|$  and  $|2\pi - |\phi_1 - \phi_2||$  is chosen ( $\Delta\phi$ ),
- the invariant mass of a system formed by the three jets with the highest transverse momentum ( $m(3j)$ ),
- the invariant mass of a system formed by the four jets with the highest transverse momentum, the lepton and the missing transverse momentum ( $m(4j, l, \nu)$ ),
- the pseudorapidity of a system formed by the four jets with the highest transverse momentum, the lepton and the missing transverse momentum ( $\eta(4j, l, \nu)$ )

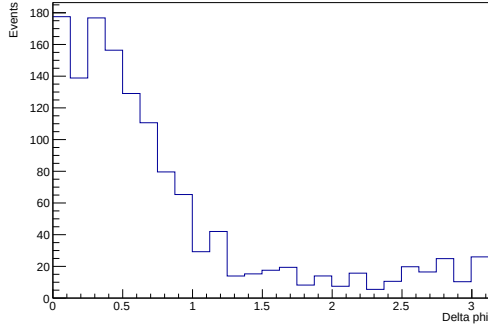
are considered and histograms are created for simulated  $t\bar{t}$ - and  $Z'(1000 \text{ GeV})$ -events. Those histograms are shown in Figure 7. The discriminants  $\Delta\phi$ ,  $m(3j)$  and  $\eta(4j, l, \nu)$  have similar distributions for both samples where the main difference are the larger fluctuations for the  $Z'$ -sample due to the smaller number of selected events. The distribution for the discriminant  $p_{T, \text{miss}}$  is fallig for both samples while the decrease is exponential for the  $t\bar{t}$ -sample, no function describing the decrease of the  $Z'$ -sample is obvious. In the distribution 4, both samples show a peak at different masses. The peak for the  $t\bar{t}$  sample is around  $\approx 500 \text{ GeV}$  which is larger than the invariant mass of two top quarks of  $\approx 346 \text{ GeV}$ , hinting towards problems of the Monte Carlo model, while the peak of the  $Z'$ -sample is located at around  $\approx 1100 \text{ GeV}$  which is close to the mass of the hypothesised particle. Since the discriminant  $m(4j, l, \nu)$  should roughly correspond to the invariant mass of the final state of the decay considered, a peak would expect in the distribution at the mass of a hypothesised  $Z'$ . Due to this clear signature of a possible signal, the discriminant  $m(4j, l, \nu)$  is chosen as the final discriminant in this analysis.



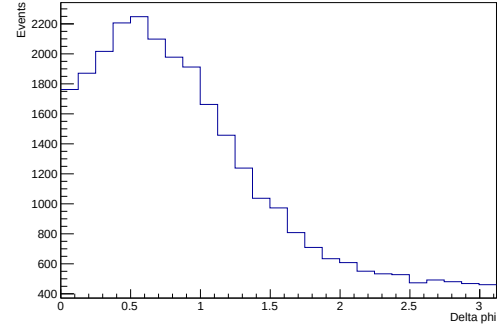
(a) Missing transverse momentum in each event for  $Z'(1000 \text{ GeV})$ .



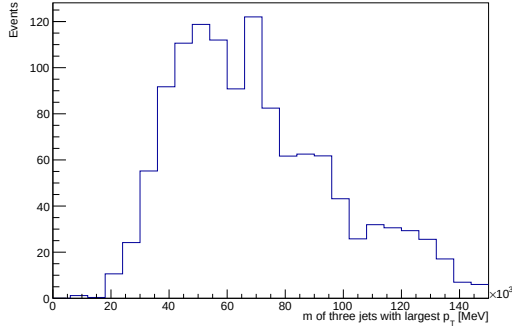
(b) Missing transverse momentum in each event for  $t\bar{t}$ .



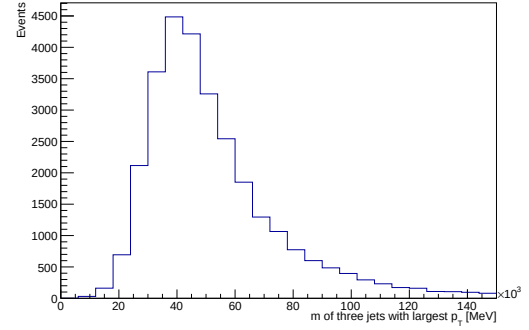
(c) The difference in the azimuthal angle of the missing transverse momentum and the lepton for  $Z'(1000 \text{ GeV})$ .



(d) The difference in the azimuthal angle of the missing transverse momentum and the lepton for  $t\bar{t}$ .

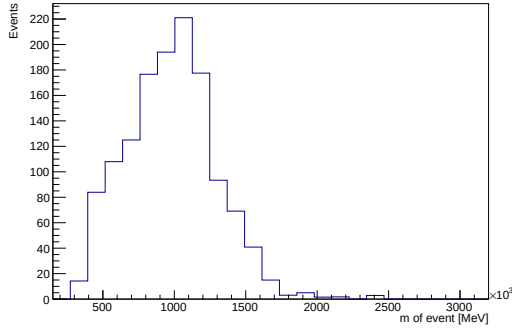


(e) Invariant mass of the three jets with the highest transverse momentum for  $Z'(1000 \text{ GeV})$ .

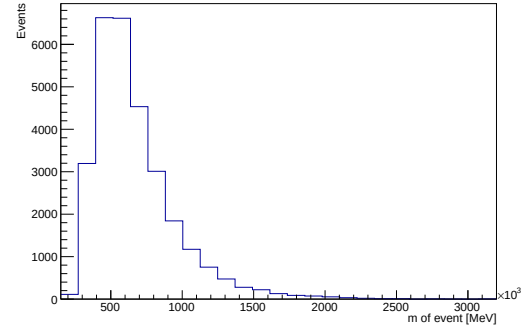


(f) Invariant mass of the three jets with the highest transverse momentum for  $t\bar{t}$ .

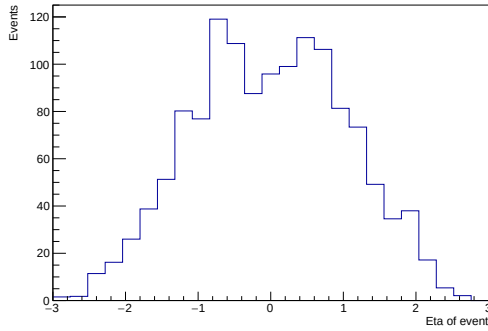
**Abbildung 7:** Histograms of the first three discriminants investigated for simulated  $Z'(1000 \text{ GeV})$ - and  $t\bar{t}$ -events where the lepton is an electron.



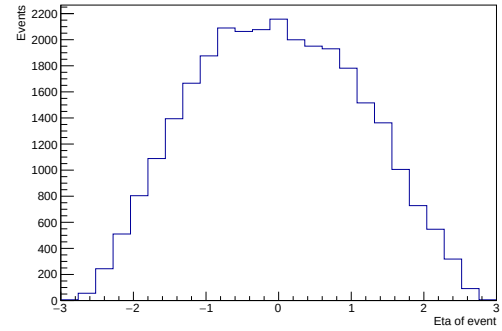
(a) Reconstructed invariant mass of the four jets with the highest transverse momentum, the lepton, and missing transverse momentum for  $Z'(1000 \text{ GeV})$ .



(b) Reconstructed invariant mass of the four jets with the highest transverse momentum, the lepton, and missing transverse momentum for  $t\bar{t}$ .



(c) Pseudorapidity of a system formed by the four jets with the highest transverse momentum, the lepton, and the missing transverse momentum for  $Z'(1000 \text{ GeV})$ .



(d) Pseudorapidity of a system formed by the four jets with the highest transverse momentum, the lepton, and the missing transverse momentum for  $t\bar{t}$ .

**Abbildung 8:** Histograms of the fifth discriminant investigated for simulated  $Z'(1000 \text{ GeV})$ - and  $t\bar{t}$ -events where the lepton is an electron.

## 5.5 Agreement of simulation and data

To test the agreement of the simulated samples and the data samples the number of expected events  $N = w\mathcal{L}\sigma(A \cdot \epsilon)$  is calculated, where  $\mathcal{L} = 1000 \text{ pb}^{-1}$  is the integrated luminosity of the data sample used,  $\sigma$  is the cross section of the considered process listed in Table 3, the weight factor  $w = \mathcal{L}\sigma/N_{\text{MC}}$  where  $N_{\text{MC}}$  is the number of events in the simulated samples listed in Table 3 and  $(A \cdot \epsilon)$  is the efficiency of the selection listed in Table 2. For data, events with an electron were used for this part of the analysis. The considered data sample contains 1112 events after the selection while the number of expected events of the background processes is 716. The number of expected events, where the determined values for each process are shown in Table 3, is lower than the number of observed events. To further investigate discrepancies between the data and

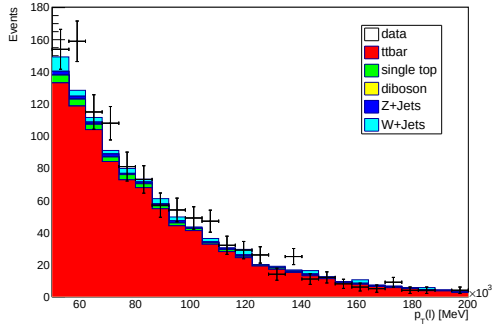
simulated events, the distributions of the quantities from Section 5.3 and the chosen final discriminant are created for the data sample and the simulated processes and shown in Figure 9 and Figure 10. In these plots, all simulated events are weighted with the weight factor shown in Table 3, to normalize these to their different cross sections.

All the distributions in Figure 9 show very similar shapes for the data sample and simulated samples, but the excess of events in the data sample, that was already observed in the number of total events, is also visible. The same is true for the final discriminant. Since the observed excess in the final discriminant is not concentrated in a peak, as it would be expected for the products from the decay of a hypothetical particle, this excess is not initially interpreted as a signal from a new particle but rather shows the imperfections of the simulated events.

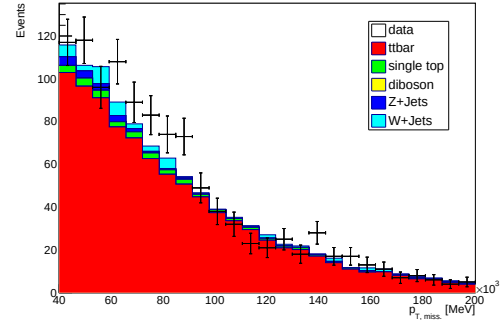
**Tabelle 3:** Number of simulated events  $N_{\text{MC}}$ , the expected cross section  $\sigma$ , determined weights  $w$  and the determined number of expected events for all Monte Carlo samples which include an electron  $N_{\text{expected}}$ .

| sample     | $N_{\text{MC}}$ | $\sigma$ / pb | $w$ / $10^{-2}$ | $N_{\text{expected}}$ |
|------------|-----------------|---------------|-----------------|-----------------------|
| ttbar      | 7847944         | 252.82        | 3.221           | 411                   |
| singletop  | 1468942         | 52.47         | 3.572           | 15                    |
| diboson    | 922521          | 29.41         | 3.188           | 0.1                   |
| zjets      | 37422926        | 2516.2        | 6.724           | 9                     |
| wjets      | 66536222        | 36214         | 54.427          | 280                   |
| zprime400  | 100000          | 110           | 110.000         | 2735                  |
| zprime500  | 100000          | 82            | 82.000          | 3379                  |
| zprime750  | 100000          | 20            | 20.000          | 486                   |
| zprime1000 | 100000          | 5.5           | 5.500           | 54                    |
| zprime1250 | 100000          | 1.9           | 1.900           | 7                     |
| zprime1750 | 100000          | 0.3           | 0.300           | 0.2                   |
| zprime2000 | 100000          | 0.14          | 0.140           | 0.03                  |
| zprime2250 | 100000          | 0.067         | 0.067           | 0.008                 |
| zprime2500 | 100000          | 0.035         | 0.035           | 0.002                 |
| zprime3000 | 100000          | 0.012         | 0.012           | 0.00002               |

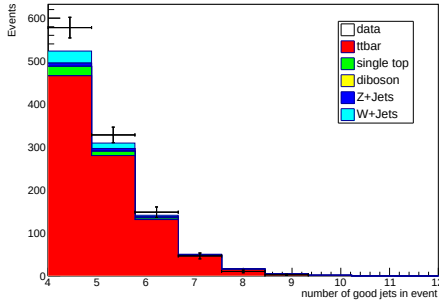




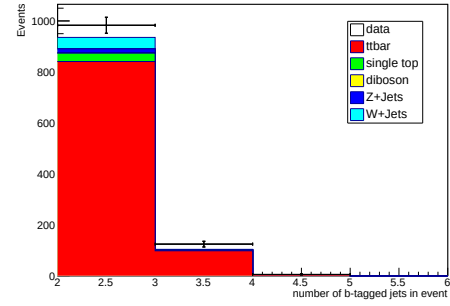
(a) Transverse momentum of the lepton.



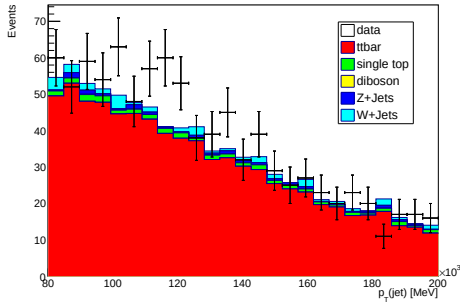
(b) Missing transverse momentum.



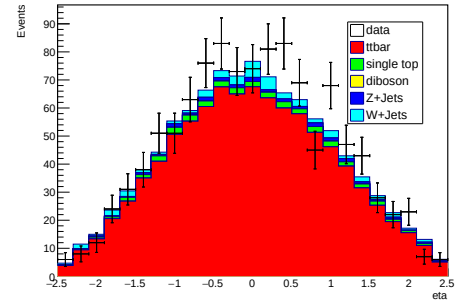
(c) Number of good jets.



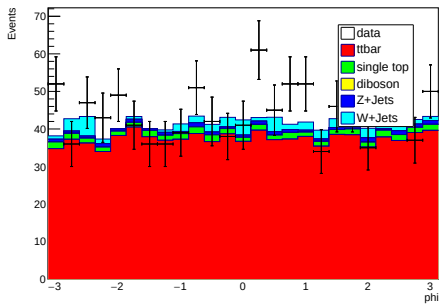
(d) Number of b-tagged jets.



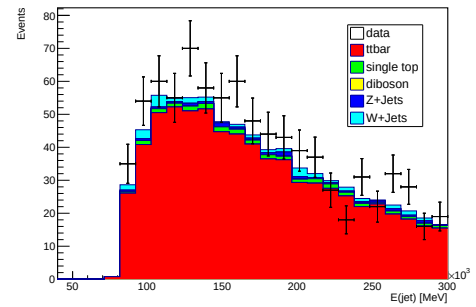
(e) Transverse momentum of jet with highest transverse momentum in each event.



(f) Pseudorapidity of jet with highest transverse momentum in each event.

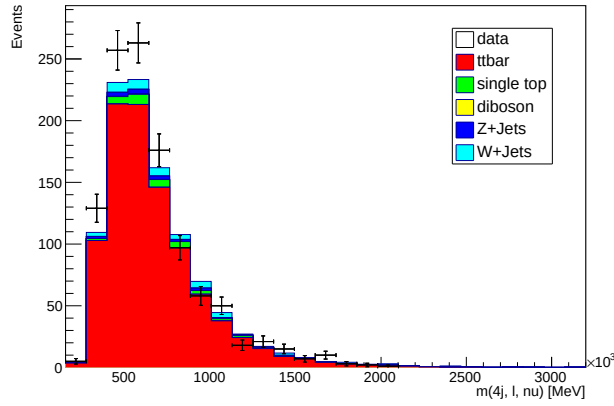


(g) Azimuthal angle of jet with highest transverse momentum in each event.



(h) Energy of jet with highest transverse momentum in each event.

**Abbildung 9:** Stacked plots of the fundamental distributions investigated in Section 5.3 with all simulated SM backgrounds which include an electron.



**Abbildung 10:** Stacked plot of the discriminant  $m(4j, l, \nu)$  with all simulated SM backgrounds which include an electron.

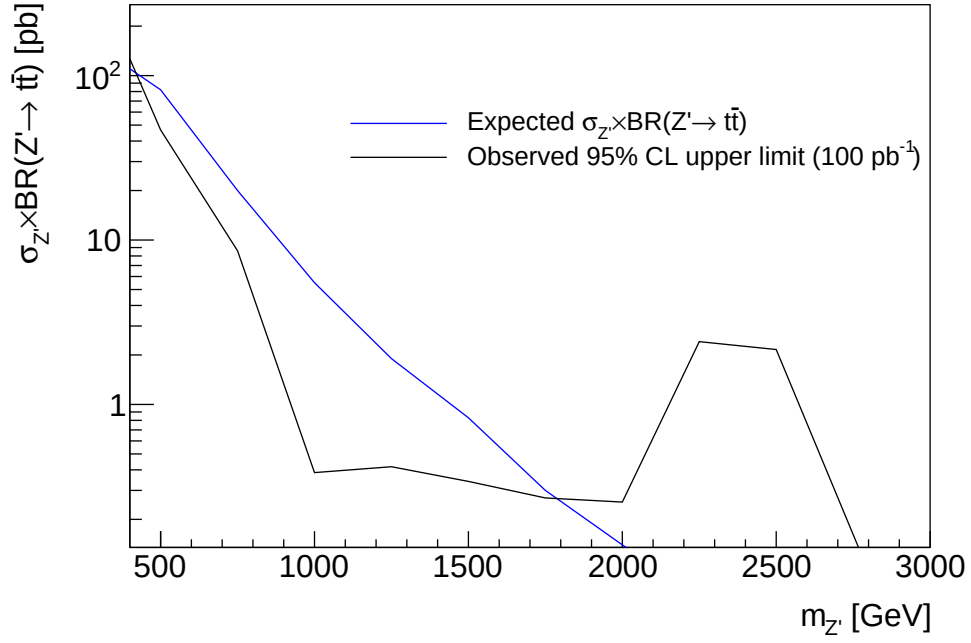
## 5.6 Statistical analysis

To investigate the agreement of the simulated samples and the data samples quantitatively, the  $\chi^2$  value for the distribution of the data and the sum of the simulated background events is determined for the chosen final discriminant. Furthermore, the  $p$ -value is determined for 25 degrees of freedom, corresponding to the 25 bins in the histograms. The values are calculated as

$$\chi^2 = 32.9652$$

$$p = 0.131945$$

and therefore, larger than the limit for claiming evidence for a signal of  $2.7 \cdot 10^{-3}$ . To further investigate the possibility of a  $Z'$  signal in the data, the 95% exclusion limit for the different mass hypotheses is determined by varying the cross section with a parameter  $s$  and replacing the cross section in the weights to be  $\sigma' = s\sigma$  until the  $p$ -value between the sum of the simulated SM processes and  $Z'$  and the data in the final discriminant is  $\approx 0.05$ . This value corresponds to a probability of  $\approx 5\%$  to observe a signal-like background fluctuation like this by chance, allowing to reject the background-only hypothesis with a probability of 5% to do so although it was such a statistical fluctuation. The values found this way for the parameter  $s$  and the new cross section for the different mass hypotheses are listed in Table 4. Furthermore the cross sections  $\sigma'$  at the 95% confidence level (CL) and the predicted cross sections are plotted against the mass of  $Z'$  in Figure 11. Areas of invariant mass where the observed cross section at 95% CL is smaller than the theoretical cross section can be excluded for the mass of a hypothetical  $Z'$ , because otherwise it would be expected to be observed already. Therefore a mass range from  $\approx 410$  GeV to  $\approx 1800$  GeV can be excluded in this analysis.



**Abbildung 11:** Histogram of the transverse momentum of the leptons in from one of the data samples created with the **TBrowser** environment from **Root**.

| sample     | $\chi^2_{25}$ | $s$       | $\sigma' / \text{pb}$ |
|------------|---------------|-----------|-----------------------|
| zprime400  | 37.7281       | 1.15      | 126.5                 |
| zprime500  | 37.0327       | 0.57      | 46.74                 |
| zprime750  | 37.5158       | 0.430001  | 8.60001               |
| zprime1000 | 37.0651       | 0.0700007 | 0.385004              |
| zprime1250 | 37.6404       | 0.220001  | 0.418001              |
| zprime1500 | 37.4218       | 0.410001  | 0.3403                |
| zprime1750 | 37.626        | 0.9       | 0.27                  |
| zprime2000 | 37.6757       | 1.82      | 0.2548                |
| zprime2250 | 37.6524       | 35.97     | 2.40999               |
| zprime2500 | 37.4851       | 61.6557   | 2.15795               |
| zprime3000 | 32.981        | 1.01      | 0.01212               |

**Tabelle 4:** The parameter  $s$ , the  $\chi^2$ -value und the modified cross section for all  $Z'$  samples at the 95% confidence limit.

## 6 Discussion

In this analysis, a search was conducted for a hypothetical  $Z'$ . In the process, multiple fundamental distributions of events have been visualised and compared with the expected

simulated background from the SM processes. It was found, that there is an overall excess of events in data compared to the simulated events, which cannot immediately be interpreted as a signal of a new particle, but rather as imperfections in the MC simulations. One other interpretation is that the excess could be combinatorial background that was not removed by the cut. To account for that the analysis could be done again with more restrictive cuts. Finally, a statistical analysis of simulated events and data was conducted and no evidence for a signal was observed. Since no signal was observed, limits on the cross section of the hypothetical  $Z'$  have been determined for the considered masses at 95% CL. A mass range from  $\approx 410$  GeV to  $\approx 1800$  GeV has been excluded for a hypothetical  $Z'$ .

Results of the analysis could most likely be improved by optimising the used cuts and utilising the muon samples in addition to the electron samples.

## Literature

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